Milestone 2 · Architecture and plan

Transylvanian Folk & Rap Dual-Mode Creative Agent

Poezie ardelenească & rap ardelenesc · fine-tuned SLM + RAG + MCP tools

Team: Platinum Records

Goal and what we commit to

Recap of the RFC, plus the choices that keep us on the grade-10 track

Problem

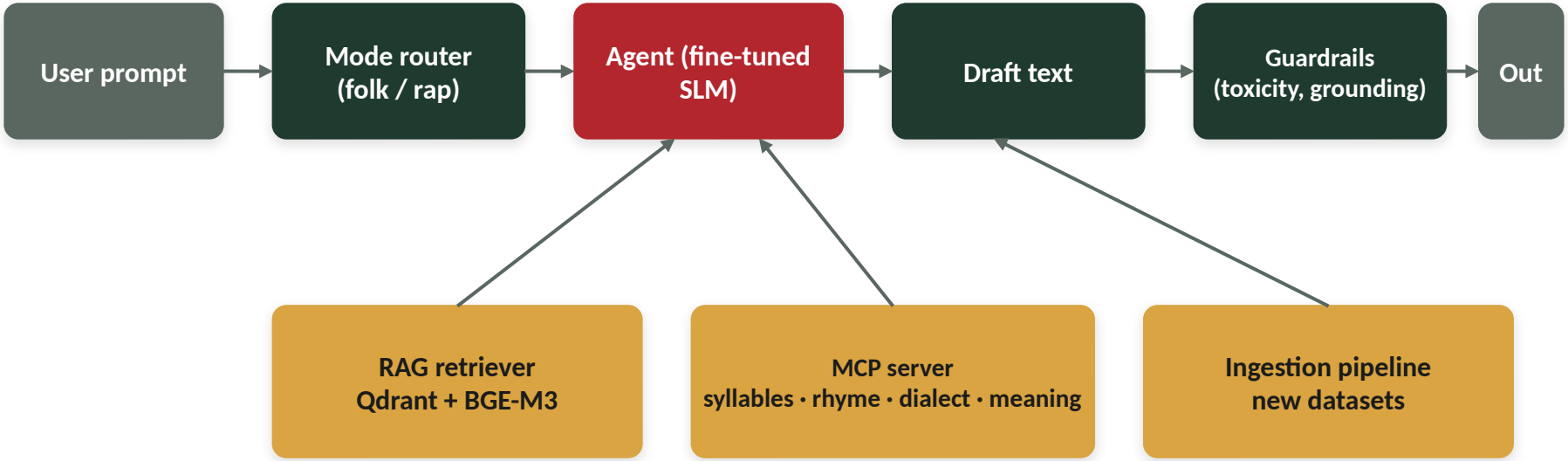
- LLMs write fluent Romanian, but Transylvanian dialect, idioms and poetic structure come out inconsistent
- Templates enforce structure but sound repetitive
- Task: input = prompt + mode (folk or rap); output = poem or lyrics that keep the meaning, with regional words, rhyme and rhythm

Commitments for a 10

- Fine-tuned SLM, not a commercial API
- Agent with 6+ MCP tools that drive decisions
- RAG over a curated regional knowledge base
- New-dataset ingestion, task-specific fine-tuning
- Evaluation + toxicity/hallucination guard
- Commercial model kept only as a baseline

System architecture

Agent loop: decide, retrieve, generate, verify with tools, revise



If a check fails, the agent revises the draft (max 3 loops).

Worked example: a rap request, step by step

Illustrative trace of one request through the agent

1. **Prompt**
   „rap despre dor de casă” (rap about homesickness)

2. **Route**
   The mode router picks rap. Defaults: 4 lines, 8 syllables, scheme AABB

3. **Retrieve**
   Filter mode = rap: a style chunk (AABB, 8 syllables, theme dor) + the dialect entry bade = address for an older man

4. **Generate**
   The fine-tuned SLM writes draft v1 from the prompt, the style chunk and the dialect entry

Draft v1

Plec din Ardeal cu dorul greu

Dar casa bate-n pieptul meu

Iar timpul trece ca un zbor

Și mă-ntorc iar la casa mea, bade

Next: the checks run on these four lines

When a check fails: revise and re-check

Same request: two of four checks fail on line 4

Tool	Result on draft v1	Verdict
`syllable_counter`	[8, 8, 8, 10]: line 4 has 10 syllables, target is 8	FAIL
`rhyme_checker`	Scheme found A A B C: „bade” does not rhyme with „zbor”	FAIL
`dialect_lookup`	„bade” is found in the knowledge base	PASS
`meaning_check`	Draft still matches the prompt (homesickness)	PASS

Revise (loop 1 of 3)

- Rewrite only line 4: 8 syllables, end in -or, keep „bade”

→

Draft v2, line 4

Mă-ntorc, bade, că-mi e dor

8 syllables, rhymes with „zbor”

→

Re-check: all pass

- [8, 8, 8, 8], scheme AABB
- Guardrails pass; text is returned

If a check still fails after 3 loops, the agent returns the best draft with a warning instead of looping forever.

Dataset selection

Candidates below; licensing checked before use

Knowledge base (RAG)

- Folk collections by Transylvanian collectors, e.g. Pop-Reteganul
- Public-domain poets: Coșbuc, Blaga, Goga
- Dialect lexicon and idiom lists (dexonline, regional glossaries)

Fine-tuning set

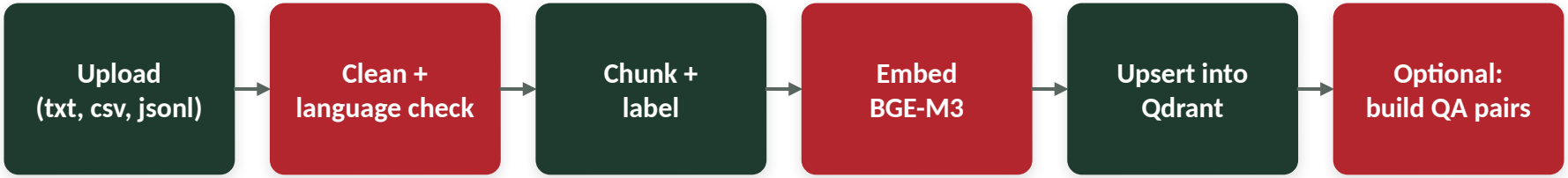
- Prompt-to-poem and prompt-to-lyrics pairs
- Synthetic pairs from a commercial LLM, human-reviewed
- Rap: own-written or licensed lyrics only; style features, not copied text

Cleaning and labels

- Diacritics normalised, duplicates removed
- Labels: mode, theme, rhyme scheme, region
- Held-out test set of 100 prompts, never used in training

Ingesting new datasets

The user can add a corpus without touching code



Why it matters

- Required by the project: ingestion of new datasets
- Same pipeline fills the initial knowledge base

Interface

- CLI command and a tool exposed through MCP
- Reports chunk count and rejected rows

Chunking strategy

Chunk by the unit of meaning, not by token count

Source	Chunk unit	Metadata
Folk poems	One stanza, 1-stanza overlap	author, region, theme, rhyme scheme
Rap lyrics	4-8 line verse (bar group)	artist style tags, tempo, themes
Dialect lexicon	One entry (word + meaning + example)	word type, region, register
Idioms / sayings	One saying with its explanation	topic, tone

Why: a stanza, verse or entry is the smallest unit that keeps meaning and rhyme intact. Metadata lets retrieval filter by mode (folk or rap) before ranking.

Vector database and embeddings

Decision: Qdrant + BGE-M3

Option	Pros	Cons
Qdrant (chosen)	Metadata filters, hybrid search, easy local Docker	One more service to run
Chroma	Simplest setup	Weaker hybrid and filtering
FAISS	Fast, no server	No metadata store of its own

Embedding model

- BGE-M3: multilingual, handles Romanian, supports dense and sparse vectors for hybrid search
- Fallback to multilingual-e5 if memory is a problem; compared on a retrieval test set

Model choices

Three systems compared on the same test set

A. Commercial baseline

- Prompted LLM via API
- Reference only: caps grade at 8
- Shows what fine-tuning must beat
- Why: measures the gap we need to close

B. SLM, not tuned

- Base 3-4B model (Qwen / Gemma / RoLlama candidates)
- Same RAG and tools
- Caps grade at 9
- Why: isolates what RAG and tools add

C. SLM, fine-tuned

- Same base, QLoRA on our data
- Final system
- Allows grade 10
- Why: adapts style and dialect; runs on modest hardware

How we pick the SLM

Final choice made in week 7 from a small, fair test

Selection criteria

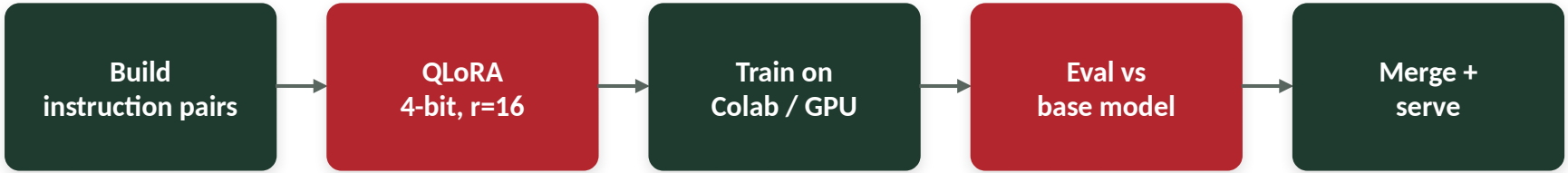
- Romanian quality: human rating on 10 prompts
- Fits a free GPU in 4-bit for QLoRA
- Permissive licence
- Supports tool / function calling
- Small enough to be fast (about 4B or less preferred)

Process

- Candidates: Qwen, Gemma, a Romanian-tuned Llama (RoLlama)
- Same 10 prompts, with and without retrieved context
- Score each; keep one main model and one backup
- Record the decision in docs/model_choice.md
- Fallback: a smaller model if training does not fit

Fine-tuning plan

Task-specific tuning for both modes



Training tasks

- Prompt to folk poem
- Prompt to rap verse
- Dialect Q&A (meaning of regional words)
- Tool-call format for the agent

Safeguards

- Hold-out set, no leakage from train
- Compare perplexity and judged quality
- Fallback: prompt-tuned SLM if training fails

RAG strategies to try

Each one is tested on the same retrieval and quality benchmarks

Strategy	Idea	Expected effect
Mode-filtered retrieval	Filter chunks by folk or rap	Less style mixing
Hybrid search	Dense + sparse (BM25) scores	Catches rare dialect words
Re-ranking	Cross-encoder reorders top-20	Better top-3 relevance
Query rewriting	SLM expands the prompt with regional terms	Better recall
Two-pass retrieval	Vocabulary first, style examples second	Richer context

What the agent decides

Four decisions per request; each one uses a tool or the retrieved context

1. Which mode?

- Folk, rap, or unclear (ask) / unsafe (refuse)
- Input: prompt text; rule: keywords + SLM classification

2. What to retrieve?

- Which filter (mode, theme) and how many chunks
- Skip retrieval if the prompt needs no regional detail

3. Which check to run?

- Syllables, rhyme, dialect lookup, meaning check
- Chosen per draft, based on mode and what is missing

4. Accept or revise?

- If a check fails, rewrite with the error (max 3 loops)
- If toxicity or ungrounded claims remain, refuse or soften

MCP server and agent tools

The agent decides which tool to call at each step

`syllable_counter`

Counts syllables per line

`rhyme_checker`

Checks rhyme scheme and quality

`dialect_lookup`

Finds regional words and meaning

`retrieve_context`

RAG search with mode filter

`meaning_check`

Compares output vs. prompt intent

`ingest_dataset`

Adds a new corpus

Exposed by a standalone MCP server (Python SDK, stdio). Each tool has a typed schema, so the agent discovers and calls it, and any MCP client can reuse it.

Golden dataset

About 100 questions with expected answers, frozen in week 8

How we build it

- 3 task types: dialect Q&A (exact answer), retrieval (expected chunk ids), generation (constraints + one reference)
- ~100 items: 40 dialect/retrieval, 30 folk, 30 rap
- Written by the team, checked by 2 native Transylvanian speakers
- Never overlaps with the knowledge base or training pairs
- Dev split for tuning, locked test split (30%) for final results
- File: data/eval/golden_v1.jsonl

Example item

```
{
 "id": "g-017",
 "task": "generation",
 "mode": "folk",
 "prompt": "poezie despre
   mama care coace pâine",
 "constraints": {
   "scheme": "ABAB",
   "syllables": 8,
   "dialect_terms_min": 2 },
 "expected_chunks": ["folk-231"],
 "reference": "(verse by team)"
}
```

Evaluation, toxicity and hallucinations

Measured, not assumed

Structure

- Syllable accuracy per line
- Rhyme scheme match rate
- Both measured automatically by our MCP tools

Quality

- Dialect authenticity, rated by native speakers (1-5)
- Meaning preservation, LLM judge + human sample
- Systems A, B, C compared

Safety

- Toxicity filter on every output, regenerate if flagged
- Grounding check: dialect claims must match retrieved entries
- Refuse hateful or offensive requests

Timeline, part 1: weeks 6 to 9

Week 6 is this presentation only; real work starts in week 7

Week	Work	Experiment / output
6 (now)	Architecture presentation. Upload diagrams and plan to docs/ on GitHub. No implementation yet.	Deliverable: slides + docs file
7	Ingestion + cleaning, Qdrant running, golden dataset draft (30 items), syllable_counter + MCP skeleton, GPU access, SLM selection test.	E1: embeddings BGE-M3 vs multilingual-e5
8	Baseline RAG with untuned SLM, rhyme_checker + retrieve_context tools, golden dataset v1 frozen (~100 items).	E2: chunking by stanza vs fixed size
9	Progress check: demo of the agent loop (RAG + 3 tools); commercial baseline (A) and untuned SLM (B) scored.	E3: RAG strategies (filter, hybrid, re-rank)

Every row is a commitment: each week ends with visible commits and results in the repo.

Timeline, part 2: weeks 10 to 14

Fine-tuning, full evaluation, and the final demo

Week	Work	Experiment / output
10	Fine-tuning pairs built, first QLoRA run, ingest_dataset tool, toxicity filter v1.	E4: QLoRA run 1 vs base on dev split
11	Second QLoRA run, dialect_lookup + meaning_check tools, grounding check against hallucinations.	E5: systems A vs B vs C on full golden set
12	GitHub review before Christmas: ingestion, RAG, 6 tools, tuned model v1 and results v1 all committed, README with run steps.	E6: agent with vs without revision loop
13	Fix top error categories, human evaluation with native speakers, documentation, code freeze.	E7: human rating of dialect authenticity (1-5)
14	Final presentation: 15 min demo + documentation + source code.	Final results table

Weeks 12 and 14 are fixed checkpoints; weeks 8-9 are the labs progress check.

Team roles

5 students, roles by skill

Data lead	Corpus, cleaning, licensing, labels
RAG lead	Chunking, Qdrant, retrieval experiments
Fine-tuning lead	QLoRA, training runs, model comparison
MCP / agent lead	Tools, agent loop, guardrails
Eval + docs lead	Test set, metrics, demo, documentation

Project requirements, covered

- Agent with multiple tools (6 MCP tools)
- Support ingestion of new datasets
- Task-specific fine-tuning (QLoRA)
- RAG with strategies compared
- MCP server for our tools
- Golden dataset + evaluation on 3 systems
- Toxicity / hallucination handling
- Fine-tuned SLM: grade 10 tier

Originality: dual-mode creative agent where tools check structure and retrieval grounds dialect.